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Life-Cycle Labor Supply and Fertility: Causal Inferences from Household Models

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Although estimates of the fertility–labor supply relationship abound, a full appreciation of the interpretation of such estimates has been lacking, regardless of the empirical strategy employed. This paper attempts to elucidate, within the context of a life-cycle decision-making process, the information contained in the estimated association between fertility and labor supply as calculated from “single” and “simultaneous-equations” estimation techniques. We also present a statistical methodology based upon the occurrence of twins in the first pregnancy and provide estimates, using that methodology, of the extent to which women’s life-cycle labor supply decisions respond to exogenous (and, in this case, unanticipated) extra children.

I. Introduction

The purpose of this paper is to explore the theoretical and empirical relationship between female labor supply and fertility in the context of a life-cycle decision-making process. Our starting point is the well-documented empirical finding that younger women with lower measured fertility engage in market activity more frequently and more intensively, as estimated from participation or hours regression equations that treat fertility as exogenous (Gronau 1973; Heckman 1974; Heckman and Willis 1977). This research stands in sharp con-

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trast to an equally vast literature that treats fertility as a choice variable (Becker and Lewis 1973; Willis 1973). Recognition of this inherent conflict has spurred several attempts at specifying and estimating simultaneous-equations models (Cain and Dooley 1976; Schultz 1977; Fleisher and Rhodes 1979). Although estimates of the fertility–labor supply relationship abound, a full appreciation of the interpretation of such estimates has been lacking, regardless of the empirical strategy.¹ This paper attempts to elucidate, therefore, what information is contained in the estimated association between fertility and labor supply as calculated from “single” and “simultaneous-equations” estimation techniques. In addition, an empirical methodology is developed to estimate the impact of exogenously distributed children on life-cycle labor supply decisions.

In Section II, a multiperiod utility-maximization model is presented which is intended to illuminate the essential links between empirical estimates of the fertility–labor supply association and economic theory. The model incorporates, within a static framework, notions of child-specific investments and market-specific human capital atrophy. In Section III, we present a statistical methodology based upon the occurrence of twins in the first pregnancy and provide estimates in Section IV, using that methodology, of the extent to which women respond to exogenous extra children. Section V summarizes.

II. Fertility and Life-Cycle Labor Supply:

A Theoretical Framework

In this section we formulate a multiperiod optimization model which incorporates two notions central to the fertility and life-cycle labor supply literatures—a production function for child services and an intertemporal relationship between earnings capacity and labor supply. We first use the model to elucidate the difference in information provided by the population association between fertility and labor supply, that obtained from the conceptual experiment in which children are distributed randomly across the population, and that provided by simultaneous-equations estimates based on household data. We then show how the results of the experiment, which are generated in Section III, can be used to test the implications of the specific model and its structure.

A model which would capture many of the important aspects of the

¹ A notable exception to these empirical labor supply traditions is Mincer (1963), in which the inappropriateness of including a fertility variable among the set of labor supply regressors is suggested. This paper is in the spirit of that seminal article, providing some possible theoretical rationales for such specifications and their interpretation.

relationship between fertility and labor supply would distinguish at least three periods—two childbearing periods, early and late, in which child services are produced with home time, and a third period corresponding to a time when children are mature and child investment is minimal. Such a model could thus capture optimal spacing decisions as well as the costs of changing contraceptive strategies associated with such behavior. To reduce complexity, however, we initially ignore this latter complication, which is discussed below, and instead for illustrative purposes construct a two-period model, corresponding to the fertile and nonfertile (no child services produced) periods of the life cycle. The utility function is given by

$$U = U(n, l_1, l_2, X, q; \epsilon), \quad (1)$$

where l_i is home time in period i , n is the number of children, q is the services (quality) of each child, and X represents other goods consumed. Function (1) has the usual neoclassical properties, but the population is heterogeneous with respect to preference orderings over the five goods, as represented by a randomly distributed shift parameter ϵ , such that $U_{l_i\epsilon}, U_{n\epsilon} \neq 0, U_{x\epsilon}, U_{q\epsilon} = 0$.² The production of child services q is assumed to occur in the first period and is described by a production function (2). While for simplicity it is assumed that q is produced jointly with l_1 and n , function (2) captures the

$$q = \Theta(l_1, n) \quad (2)$$

critical assumptions found in the household production fertility literature (Becker and Lewis 1973; Willis 1973)—nonjointness of quality production across children and time intensity—since we assume that $\Theta_{l_1} > 0$ and $\Theta_n < 0$ and that no purchased-goods inputs are used to produce q . The individual begins the planning period with an exogenously given stock of human capital H_1 , which then depreciates as a function of home time in period 1. The dependence of H_2 on l_1 , and thus the price of l_2 and l_1 , is given by (3) and will be referred to as serial dependence:³

$$H_2 = \psi(l_1) \psi' < 0. \quad (3)$$

The lifetime full income constraint, assuming a zero rate of interest, is thus

² These parameterizations of preferences are meant to be illustrative. Relegation of unobserved differences in preferences to additive error terms in linear demand or “structural” equations, as is common, implies as well a particular, although less easily interpretable, structure to “tastes” heterogeneity.

³ Alternatively, the appreciation of human capital, or, more generally, a threshold function in which home time above some value l_1^* leads to depreciation and below l_1^* induces a rise in the stock of human capital, can be introduced into the multiperiod model. The main points of the subsequent analysis would be unaffected.

$$F = r(H_1T_1 + H_2T_2) + A = p_n n + p_x X + r(H_1l_1 + H_2l_2), \quad (4)$$

where the prices for X , p_x , and n , p_n , and the return to the human capital stock, r , are fixed and known in advance with certainty, A is exogenously given asset income, and the T_i represent full time in each period. Note that p_n is the cost of increasing the number of children independent of the level of q chosen, for example, birth and contraceptive costs, and is not the shadow price of child services.⁴ The model implicitly yields four demand functions of conventional form, in which each of the four observable control variables l_1 , l_2 , n , and X is a function of the exogenous variables p_n , p_x , rH_1 , F , and ϵ .

a) *Heterogeneity, Fertility Effects, and
"Structural" Equation Estimation*

To compare the informational content of the correlation between fertility variables and labor supply in a heterogeneous (in ϵ) population with that generated from the same world but in which fertility outcomes are generated exogenously, assume that F , r , H , and p_x are "controlled," but we have no information on p_n or the unobservable tastes parameter ϵ , both of which may vary across individuals. An approximation to the slope of the relationship between l_1 and n in the population, assuming interior solutions, is given by

$$\frac{dl_1}{dn} = \left[\frac{\phi_{21}}{\phi_{11}} + \frac{(U_{n\epsilon}\phi_{12} + U_{l_1\epsilon}\phi_{22} + U_{l_2\epsilon}\phi_{23})d\epsilon}{\phi_{11}dp_n} \right] \left[\frac{(U_{n\epsilon}\phi_{11} + U_{l_1\epsilon}\phi_{21} + U_{l_2\epsilon}\phi_{31})}{\phi_{11}dp_n} \right]^{-1}, \quad (5)$$

where ϕ is the determinant of the bordered Hessian and the ϕ_{ij} are the cofactors from the i th row and j th column of the Hessian determinant. As is well known, all of the predictions of the model, the tastes-constant price-effect information, are contained in the cofactor terms; that is, $\phi < 0$, $\phi_{ii} > 0$, etc. *Thus only empirical estimates which provide information on the ϕ_{ij} can be used to test the model or to ascertain the potential effects of policy on behavior.* As can be seen, however, (5) reflects both the structure of the model as given by the ϕ_{ij} terms and the unobserved population difference in ϵ and p_n . If it could be assumed that preferences did not vary ($d\epsilon = 0$), however, the slope of the fertility-labor supply relationship would provide information on (compensated) price effects, being equal to ϕ_{21}/ϕ_{11} , the ratio of the compensated cross-price effect of p_n on l_1 (or rH_1 on n) to the compen-

⁴ Given (2), the shadow price of child quality would be a function of both the wage (rH) and p_n .

sated own-price effect on n . Since $\phi_{11} > 0$, the sign of (5) would thus provide the sign of the effect, say, of a compensated change in contraceptive costs on labor supply during the childbearing period.

While the actual relationship between fertility and labor supply thus cannot provide information on tastes-constant cross-price effects when both p_n and ϵ vary and are unobserved, it can easily be demonstrated, based on the rationing or conditional demand literature (Tobin and Houthakker 1950–51; Pollak 1969), that treating n as a parameter $\bar{n}$ and varying it exogenously around the unconditional equilibrium that would have obtained, that is, around planned or optimal n , does yield compensated price-effect information, that is,

$$\frac{dl_i}{d\bar{n}} = \frac{\phi_{i+1,1}}{\phi_{11}} = \frac{dl_i/dp_n}{dn/dp_n}, \quad i = 1, 2. \quad (6)$$

Thus if we had an experiment in which heterogeneous (in ϵ) individuals were constrained randomly with respect to fertility—some bearing more children or some fewer than the planned or optimal number—the observed differences in labor supply associated with $\bar{n}$ would provide the tastes-constant compensated cross- and own-price-effect ratio.

These considerations illustrate the rationale for the use of simultaneous-equations techniques for estimating the effect of n on labor supply *as well as their redundancy*. Use of two-stage least squares, say, is an attempt to obtain an estimate of dl_i/dn where n does not reflect variation in unobserved preferences—which it otherwise will, even if tastes differ in the population only with respect to preferences for l_i (i.e., $U_{l_i\epsilon} \neq 0$ only). However, the only source of variation in n across individuals not due to differences in preference orderings must be differences in the own price of children, p_n (p_x , F , r , and H , but not ϵ , are assumed to be fixed, or “controlled”). The identifying instrument for the second-stage labor supply equation conditional on “predicted” n is thus, in this case, p_n or its determinants. Of course, this normalization is perfectly arbitrary, as it is possible to use any one of the set of labor supply determinants, omitted from that relationship, as the instrument for fertility. Moreover, since the coefficient on predicted fertility in the conditional labor supply equation yields only the *ratio* of compensated p_n effects, it is obvious that the same information could be obtained by estimating the usual demand equations for n and l_i , from which Hicks-Slutsky price and income effects are obtained directly. Note also that, while simultaneous-equations techniques can yield consistent estimates of (6), the effects of other prices and income on labor supply in an equation conditioned on fertility will, of course, differ from the Hicks-Slutsky effects obtained from

the (unconditional) demand equations, which are functions only of prices and income.⁵

Thus, if preference orderings for home time vary parametrically and fertility is a choice variable whose price also varies, labor supply equations, which include some measure of actual fertility among the regressors, will provide inconsistent estimates of conditional price effects. With all prices known, use of simultaneous-equations techniques would yield consistent estimates but no more information than that provided by conventional demand equations. Conversely, if p_n is not known but varies, and preferences for home time do not differ among individuals, actual fertility may be substituted as a determinant of labor supply to yield consistent estimates of the price-effect ratios (6). In the latter two cases, however, only consistent estimates of *conditional* price effects are obtained. Estimation of labor supply equations without fertility or p_n included among the regressors, however, would obviously yield biased estimates of the Hicks-Slutsky price and income effects.

b) The Informational Content of Consistent Estimates of Fertility Effects on Life-Cycle Labor Supply

In Section III we demonstrate that consistent estimates of (6) can be obtained by using twin births without the need for information on prices or ϵ . As we have seen, such an estimate yields the signs of the compensated price effects dl_i/dp_n (or of the cofactors ϕ_{12} and ϕ_{13}), which are not predicted by the model as structured. Indeed, it can be demonstrated, even in a one-period model, that only if additional restrictions are placed on the utility function can more inferences be derived about the particular structure of the model, for example, serial dependence or the q -relationship, from the estimates of the effects of an exogenous fertility change on labor supply. We now place such a restriction on (1) to illustrate what further hypotheses generated from even the simple model constructed can be verified from knowledge of dl_i/dn . In particular, we concentrate on distinguishing the existence of the unobservable q -function and/or human

⁵ The signs of the Hicks-Slutsky and *conditional* compensated price effects are the same, however. E.g., in the case in which home times in periods 1 and 2 are perfect substitutes, the compensated "wage" (rH) effect on home time from the usual demand equation (which holds other *prices* fixed) is $dl/d(rH) = \lambda\phi'_{22}/\phi'$, while the own-price effect on home time conditioned on $n = \bar{n}$ is simply a scalar times the inverse of the compensated own-price effect on n ; i.e.,

$$\left. \frac{dl}{d(rH)} \right|_{n=\bar{n}} = \frac{-\lambda p_x^2}{\phi'_{11}} = \frac{-\lambda p_x^2}{\phi'} \left(\frac{dn}{dp_n} \right)^{-1}.$$

capital depreciation functions from the usual utility-function relationships. The restriction we impose, strong separability, allows a decomposition of the cofactors ϕ_{13} and ϕ_{12} , with either the q - or ψ -functions dropped, into additive terms corresponding to the utility function and to the auxiliary serial dependence or quality investment hypotheses. We show that under strong separability, however, the signs of the cofactors ϕ_{13} and ϕ_{12} provided by estimates of (6) can be used to test only for the existence of the q -function; the serial dependence hypothesis cannot in this case be disproved with such information.

Consider first the (exogenous) effects of n on l_1 and l_2 with the q -relationship suppressed, given by (7) and (8), separability assumed:

$$\frac{dl_1}{dn} = (\phi_{12}^c + \lambda r^2 \psi' H_2 U_{xx}) \phi_{11}^{-1} < 0, \quad (7)$$

$$\begin{aligned} \frac{dl_2}{dn} = & [\phi_{13}^c + \lambda r p_n U_{xx} \{r(T_2 - l_2) \psi' H_2 \\ & + \psi' [H_1 - (T_2 - l_2) \psi']\}] \phi_{11}^{-1} < 0, \end{aligned} \quad (8)$$

where

$$\phi_{12}^c = p_n U_{xx} U_{l_2 l_2} r (\Pi_{l_1}) > 0,$$

$$\phi_{13}^c = p_n U_{xx} U_{l_1 l_1} r H_2 > 0.$$

$$\Pi_{l_1} = \begin{cases} H_1 & \text{if } \psi' = 0, \\ H_1 - (T_2 - l_2) \psi' & \text{if } \psi' \neq 0. \end{cases}$$

The first terms in (7) and (8) are the cofactors from a four-good model similar to (1) but in which the home-produced q is not an argument. Thus ϕ_{12}^c and ϕ_{13}^c would correspond in sign to the standard Hicks-Slutsky cross-price effects, defining (without restriction) whether l_1 and n and l_2 and n are substitutes or complements. In this case, however, they are positive because of the separability assumption. The second terms in the numerators of (7) and (8) result solely from the serial dependence hypothesis and are also both positive, if the second derivative of the human capital ψ -function is "small" or negative. Thus, because it can be easily shown that the q -production equation (2) would add terms to (7) and (8) which are of ambiguous sign, the finding that an increase in the number of children *decreased* labor supply at any point in the life cycle would lead to a rejection of the (separable) intertemporal labor supply model which ignored time-intensive child services production. However, given that the serial dependence and utility-function terms are of the same sign, it is

obvious that, under the strong separability assumption, the existence of serial dependence in human capital cannot be verified.⁶

Finally, while the model formulated assumes that individuals are indifferent as to the timing of births, we can readily generalize the framework by also making children in each period imperfect substitutes in the utility and q -functions (9) and thereby capture optimal birth-spacing considerations, as in Razin (1979).⁷ *Mutatis mutandis*, the effects of exogenous increases in n in one period on fertility in subsequent periods will yield information on utility and q -function relationships between births in different stages of the life cycle.

III. Exogenous Fertility Effects: The Twins-First Methodology

We have demonstrated that, in the presence of heterogeneity, the population association between fertility and labor supply does not, in general, yield the compensated price-effect information contained in either the unconstrained or constrained demand equation setting. Lacking data on the price of children (or the proxies of the price) or on individual-specific utility-function parameters, one can generate consistent estimates of the signs of compensated cross-price effects only if fertility can be exogenously varied across the population in a manner unconnected to preference orderings. In the following sections, we develop and implement a methodology using nonexperimental data to simulate the appropriate experiment. In particular, we show how a natural event, the occurrence of a multiple birth or "twins," can be used as an instrument for exogenous fertility movements. The variable we propose, a twins outcome in the first birth, approximates the social experiment we wish to perform not only in that some families receive an unanticipated child, while others do not, but also in that the treatment and control groups are randomly selected with respect to characteristics that may be related to market

⁶ It is important to note that the difficulty of distinguishing the importance of child services production from serial dependence in the absence of direct information on (estimates of) eqq. (2) and (3) does not result solely from the fact that we may be limited only to information on the signs of the fertility effects on labor supply, which we provide in the next section. It is demonstrated in Rosenzweig and Wolpin (1979) that, if the effect of an *exogenous* change in past labor supply on current labor supply could be estimated, we could only discern (in a weak test) the existence of serial dependence both if separability is imposed and investment in children, as characterized by eq. (2), is suppressed.

⁷ Razin's model treats total fertility, intervals between births, as well as the total fertility span as control variables, with parental leisure fixed exogenously. The comparative statics of the model are not discussed, as they are not here, because of the difficulty of obtaining any verifiable predictions.

participation. It is therefore unnecessary to utilize any information on the determinants of labor supply behavior in order to determine the "true" exogenous fertility effect by this method.

To see why the occurrence of twins in the first pregnancy, "twins first," leads to the appropriate experiment, consider, instead, a comparison of women who have had twins in *any* birth and women who have not. Maintain for the moment the assumption that the probability of a twins birth is independent of parity. It is obvious that women with more births—and thus women with, on average, greater desired fertility—will be overrepresented in the sample of twins families. The labor supply of women with twins will therefore reflect in part any relationship between unobserved tastes for children and/or tastes for home time and labor supply as well as the impact of the additional unanticipated child which contains the price-effect information. Moreover, the per pregnancy probability of twins appears to rise with parity (Mittler 1971), reinforcing the positive covariance between desired family size, which may reflect differences in preference orderings, and twins occurrence.⁸

The first birth has the desirable feature that the population of women who experience twins in that birth would prefer the same completed family size as women who do not experience twins in the first birth. However, the straightforward comparison between these two groups of women is complicated somewhat because the probability of twins occurrence also rises with the age of the mother at conception (Mittler 1971), which is itself subject to choice and therefore may reflect heterogeneity in preferences.⁹ However, because (i) twins first is orthogonal to all other determinants of labor supply with the exception of age at first birth—that is, holding age at first birth constant—and (ii) age at first birth is orthogonal to all other determinants of twins-first occurrence, sufficient covariance restrictions exist to enable "consistent" estimation of the twins-first effect.

To demonstrate this, define P to be a continuous measure of labor supply or any other choice variable of the household, u the purely random or exogenous component of twins-first occurrence—that is, net of the age-at-first-birth effect—and ϵ the composite of all other

⁸ To correct for the rise in the probability of a multiple-birth event with parity, the ratio of twins to the total number of pregnancies was proposed in Rosenzweig and Wolpin (1980) as a variable capturing the random component of multiple births. The shortcomings of that approach (necessitated by data limitations) detailed in that paper are eliminated in the methodology proposed here. The data discussed below reveal the following pattern of *per pregnancy* probabilities of twinning for parities of one to six: .0076, .0110, .0144, .0168, .0178, and .0206.

⁹ The rise in the probability of a multiple birth from the first pregnancy is approximately linear within each of the age intervals we consider in the subsequent empirical analysis. The slope coefficient is $.453 \times 10^{-4}$ over the age range 15–45.

(observed and unobserved) labor supply determinants, including those related to preference orderings. Consider the following estimating equation:

$$P = \alpha u + \epsilon. \quad (9)$$

Since u and ϵ are uncorrelated, if u were known, α could be estimated by a simple linear regression and would be an unbiased estimate of the exogenous impact of an "extra child" on labor supply. Although u is not directly observable, it can be written as

$$u = TF - \beta AFB, \quad (10)$$

where TF is a dichotomous variable representing the occurrence of twins in the first birth, AFB is age at first birth, and β is the systematic biological association of AFB with TF . Note that u and AFB are independent.

If β is not known, two strategies can be employed. Either β can be estimated by some "consistent" method and α determined in a two-stage procedure using estimated values of u , or one can estimate α directly by the following equation, obtained by substituting (10) into (9):

$$P = \alpha TF - \alpha\beta AFB + \epsilon. \quad (11)$$

Estimation of (11) by ordinary least squares will yield a consistent estimate of α , the twins-first coefficient, even though AFB is presumed to be correlated with ϵ . The reason is that all TF variation is through u once AFB is held constant, and u is itself orthogonal to AFB .¹⁰ Of course, the estimate of the AFB coefficient, $\alpha\beta$, will not reflect solely the impact of "exogenous" movements in age at first birth on P , given that ϵ and AFB may not be independent. However, our sole interest is in obtaining an estimate of the TF effect which is purged of the influence of differential preferences, and that is accomplished regardless of the intervening biological relationship.

¹⁰ Let y_1, y_2 , and y_3 represent deviations from means for the variables P , TF , and AFB , respectively, with ϵ' and u' the residual deviations in (11) and (10). Then $y_1 = ay_2 + by_3 + \epsilon'$, where $y_2 = cy_3 + u'$. The least-squares estimator of a , $\hat{a}$, is thus

$$\hat{a} = [(\sum y_1 y_2)(\sum y_3^2) - (\sum y_1 y_3)(\sum y_2 y_3)] / [\sum y_2^2 \sum y_3^2 - (\sum y_2 y_3)^2]^{-1}.$$

With the large sample assumptions $\sum \epsilon u' = \sum y_3 u' = 0$ but with $\sum y_3 \epsilon' \neq 0$, we wish to show that $\hat{a} = a$. Carrying out the multiplication,

$$\sum y_1 y_2 = ac^2 \sum y_3^2 + a \sum u'^2 + bc \sum y_3^2 + c \sum y_3 \epsilon' + (b + 2ac) \sum y_3 u' + \sum u' \epsilon',$$

$$\sum y_1 y_3 = ac \sum y_3^2 + b \sum y_3^2 + \sum y_3 \epsilon' + a \sum y_3 u',$$

$$\sum y_2 y_3 = c \sum y_3^2 + \sum y_3 u',$$

$$\sum y_2^2 = c^2 \sum y_3^2 + \sum u'^2.$$

Given the covariance restrictions, $\hat{a} = a(\sum u'^2 \sum y_3^2 / \sum u'^2 \sum y_3^2) = a$.

The twins-first methodology thus enables the identification of an exogenous fertility event without any assumptions about the population distribution of preferences and can be used to draw inferences about life-cycle labor supply behavior discussed in Section II.¹¹ The occurrence of twins in the first birth is an extreme example of a “timing failure” early in the childbearing span, since two children appear simultaneously. If there is no reason to suppose that subsequent adaptation should differ in direction from less extreme failures, this natural event simulates having an “extra” child in a period in which the birth was not desired.¹²

IV. The Data and Applications

In this section we briefly describe the data and implement the twins-first methodology to estimate the causal impact of an unanticipated increase in fertility at one stage of the life cycle both on subsequent fertility behavior and on the life-cycle participation probabilities of women. We also obtain quantitative estimates of the extent to which the use of actual fertility in labor supply equations provides biased representations of the impact of an exogenous change in family size on female labor supply.

The Office of Population Research (1965) and the U.S. Department of Health, Education, and Welfare (1973) took national random samples of women containing detailed information on life-cycle pregnancy outcomes. In addition, each survey ascertains current labor force status, while the 1973 survey also has information on prior employment. In combination, there are over 15,000 women, of whom we used 12,605. The major exclusion was necessitated by the twins-first methodology itself, which requires the existence of a first birth. Thus only women having at least one child at the time of the survey were selected for the analysis. Since ages range from 15 to 44, both women who desire zero total children and those who merely wish to postpone births to ages greater than their current ones are thus excluded. Women in any age group who have no children are never captured in the analysis, and they may react differently to an “extra” child.

¹¹ It is possible that a multiple birth may not be totally unanticipated. Evidence exists that twinning is a heritable trait (Mittler 1971), although a twins background is an extremely poor predictor. In the 1970 *National Fertility Survey* described in Westoff and Ryder (1977), of the 142 families in which either parent had a twin sibling (including one in which both the husband and wife were siblings of twins), none had experienced a multiple birth.

¹² For an exogenous increase in any n_i to yield the price information discussed above, it may be necessary that planned fertility not be at biologically maximal fertility, since the occurrence of twins may in that case augment welfare.

Descriptive statistics for selected variables are displayed in table 1 for the three 10-year age intervals, 15–24, 25–34, and 35–44. In all, for the 12,605 women in the sample, there were 87 twins births in the first pregnancy. By the nature of the zero-children restriction, age at first birth will increase from the youngest to the oldest age group. A second column is therefore added for the two older age groups which restricts age at first birth to under 25 in order to facilitate the life-cycle simulation. Women who at the time of the survey were 15–24 years old must have had a first birth between those ages, while women currently aged 25–34 could have had their first birth at any age under 35. For the oldest age group a third column for age at first birth under 35 could also have been added to enable comparisons with the middle age group, but its similarity to the unrestricted group makes this addition unnecessary.

Of course, even with these corrections, the cross section can accurately depict the life cycle only if vintage is not a relevant characteris-

TABLE 1
MEANS AND STANDARD DEVIATIONS BY AGE AND AGE AT
FIRST BIRTH, POOLED SAMPLE

VARIABLE	AGE GROUP				
	15–24	25–34	25–34 (<i>AFB</i> < 25)	35–44	35–44 (<i>AFB</i> < 25)
<i>CEB</i> (children ever born)	1.728 (.966)	2.789 (1.58)	2.971 (1.59)	3.573 (2.14)	3.907 (2.21)
<i>AFB</i> (age at first birth)	18.91 (2.19)	21.12 (3.44)	20.18 (2.61)	22.27 (4.50)	20.31 (2.58)
<i>NOW</i> (current employment status; 1 = now working)	.2903 (.454)	.3620 (.480)	.3780 (.485)	.4476 (.497)	.4689 (.499)
<i>CHILD</i> (number of children under 6)	1.567 (.822)	1.110 (.980)	1.069 (1.01)	.3357 (.655)	.2894 (.619)
<i>TF</i> (proportion of first pregnancies that are twins)	.0064 (.080)	.0057 (.075)	.0055 (.074)	.0084 (.091)	.0067 (.082)
No. observations	2,487	5,241	4,531	4,871	3,722
No. twins in first births	16	30	25	41	25
Fraction of observations in 1973 sample	.678	.685	.687	.639	.673

NOTE.—Variables are the same in all tables.

tic. Moreover, since we have pooled two cross sections, women of different vintages are combined within a single age group—which will further distort the life-cycle picture if vintage effects are nonlinear. We could disentangle the vintage from the life-cycle pattern, given the two separate cross sections, but it is not our aim to directly explain movements of the variables depicted in table 1 in either dimension.¹³ Recall that the reason for pooling the two samples was to augment the number of twins-first observations to gain some precision in estimating twins-first effects. Therefore, although it is easily verified that the figures in table 1 reflect differences in vintage as well as life-cycle patterns, we cannot, given the data, very precisely discover the extent to which responsiveness to the occurrence of twins first is also governed by both phenomena. We present evidence below, however, that the pooled cross sections provide useful qualitative insights into the life-cycle response. More conclusive results can be obtained only with richer data.

a) Twins First and Subsequent Fertility

We first examine how the occurrence of twins in the first birth exogenously alters fertility patterns based on the pooled sample. Table 2 provides estimates of the impact of twins first on the number of children ever born and on the number of children currently less than 6 years of age for each 10-year age interval, with the appropriate age-at-first-birth restrictions for “synthetic” life-cycle comparisons. These estimates (and their associated standard errors) are from regression equations corresponding to (11), which include age at first birth. Given our previous discussion, the coefficients of the latter are not reported, although it is noteworthy that the inclusion of age at first birth in many cases substantially alters the twins-first impact. Table 2 reveals that the incidence of twins in the first birth substantially alters the life-cycle pattern of fertility but has only negligible impact on completed family size. If the behavior of women of different ages reasonably depicts the life-cycle patterns of a single cohort, then it would appear that women whose first birth occurs prior to age 25 and who experience a twins birth in the 15–24 age interval have one additional child in that interval, 0.65 more children in the 25–34

¹³ Table 1 is presented only to enable interpretation of the relative size of twins-first effects on fertility and labor supply. The age profiles of participation and fertility are, however, directionally accurate. In particular, market participation, where participation denotes actual employment, increases even more dramatically with age, controlling for vintage, owing to the increased participation at all ages between 1965 and 1973. On the other hand, since fertility has declined with newer cohorts (except for the oldest age group), the figures in table 1 overestimate, though not greatly, the life-cycle increase in children ever born.

TABLE 2
IMPACT OF TWINS-FIRST BIRTH ON LIFE-CYCLE FERTILITY:
SYNTHETIC COHORTS FROM THE POOLED SAMPLE

AGE AT FIRST BIRTH	AGE GROUP		
	15-24	25-34	35-44
<i>CEB:</i>			
Under 25	1.055 (.224)	.654 (.299)	.150 (.428)
Under 35	...	.631 (.259)	.312 (.312)
<i>CHILD:</i>			
Under 25	1.174 (.200)	-.331 (.299)	-.296 (.124)
Under 35	...	-.242 (.176)	-.125 (.101)

NOTE.—Standard errors in parentheses.

age interval, but ultimately only 0.15 more children.¹⁴ Thus the occurrence of twins in the first birth represents mainly a timing failure.¹⁵

The impact of twins first on the number of children under the age of 6 years, also shown in table 2, parallels that of children ever born. It is obvious that almost all children born within the 15–24 age interval would be currently under the age of 6 years, so that the twins-first effect on *CHILD* is almost identical in magnitude to its effect on *CEB*.

¹⁴ Since women within the youngest age group on average desire fewer than two children within that age interval, the fact that women with twins-first births have about three children in that interval implies that those women do not postpone a subsequent birth more (or less) than do women with a single child in the first birth.

¹⁵ Even if births are perfect substitutes across periods, an unanticipated birth early in the life cycle could alter family size if contraception cannot be adjusted costlessly. The relative inelasticity of completed family size with respect to the timing failure for women with first births prior to age 25, as revealed by table 2, would appear, therefore, to suggest that contraceptive adjustment costs are minimal. Given stability in preference orderings, evidence concerning the importance of *changes* in contraceptive costs on fertility control associated with the “contraceptive revolution” can also be obtained by comparing the completed family size effects of twins-first births for women completing fertility in 1965 with those for women completing in 1973. These results indicate a permanent effect only for the 1965 group of older women, those who were likely to have experienced twins in the first birth prior to the availability of the pill and other efficient contraceptive methods (see Westoff and Ryder 1977). Their completed family size exceeded that of nontwins women by 0.724 (SE = 0.507). The corresponding fertility effect for the 1973 group is only –0.012 (0.392). Moreover, because the fraction of completers who evidently desired only a single birth is actually smaller in the 1973 than in the 1965 cohort, the observed difference in twins effects between 1965 and 1973 cannot be due solely to the inevitable overshooting by women who experience a twins-first birth and who desire only one child. These results thus provide some indirect confirmatory evidence of the impact of the contraceptive revolution on the ability of families to control both timing and number of children.

TABLE 3
IMPACT OF TWINS-FIRST BIRTH ON LIFE-CYCLE LABOR FORCE
PARTICIPATION, SYNTHETIC COHORTS FROM THE
POOLED SAMPLE: MAXIMUM LIKELIHOOD
TRANSFORMED LOGIT COEFFICIENTS

AGE AT FIRST BIRTH	AGE GROUP		
	15-24	25-34	35-44
Under 25	-.371 (.212)	-.102 (.105)	.142 (.102)
Under 35	...	-.096 (.096)	.060 (.078)

NOTE.—Asymptotic standard errors in parentheses.

For the two older age groups, however, women with twins-first births have fewer younger children—presumably as a result of two phenomena. First, women with twins first have fewer extra children after the age of 25; second, they complete their family at an earlier age. Each of these is, of course, the outcome of an optimization process, and both together are merely descriptive rather than true explanations of the *CHILD* pattern. The point is that a twins-first birth gives rise to a different from average fertility *pattern* which may be connected to a different from average life-cycle labor force participation behavior.

b) *Twins First and Female Participation*

As was demonstrated in Section II, by examining the impact of twins first (an increase in fertility independent of ϵ) on labor supply at various points in the life cycle, we can examine how female labor supply is affected by variations in the costs of bearing (preventing) children. We can as well test some of the predictions of the multi-period labor supply model, possibly ruling out some combinations of assumptions.

Table 3 reports the impact of a twins-first birth on the dichotomous current work status variable of the mother derived from a maximum likelihood logit (*ML LOGIT*) estimation procedure based on the pooled sample.¹⁶ The estimates reported are transformed logit coefficients based on the population mean participation rate and should be interpreted in the same way as linear regression coefficients,

¹⁶ In using participation rather than a continuous measure of labor supply due to the exigencies of the data, it is necessary to assume that all the women in each age group work some time in the relevant age interval. For a fuller discussion of the relationship between Hicks-Slutsky price effects and participation estimates, see Heckman (1978).

namely, as changes in the probability of participation due to the occurrence of twins in the first birth, given age at first birth. The results indicate that, for the youngest women, the probability of working was smaller by .371 for those with twins in the first birth. This significant increase in home time early in the life cycle associated with an “extra” birth, as shown in (7), would thus lead to a rejection of the model of life-cycle labor supply under the separability assumption in which child quality, a time-intensive production activity, is not a choice variable. The twins-first impact declines to $-.102$ for the 25–34-year-olds and actually reverses in sign for the oldest women with twins, who have a .142 greater probability of participation. The *reduction* in home time as a consequence of an exogenous increase in births for older women is also consistent with the separable model which incorporates both serial dependence and time-intensive q -production. While the relatively large standard errors of these latter estimates should again draw the reader’s attention to the small samples of twins-first observations, the difference in the effects for the youngest and oldest age groups is statistically “significant” at conventional levels ($t = 2.17$).

The heightened participation of the oldest women who experienced twins births, while it is consistent with the general life-cycle model of Section II, casts doubt on the importance of human capital serial dependence. If the evident initial withdrawal from market activity induced by an extra child early in the life cycle permanently reduced the optimal human capital investment or otherwise adversely affected future wage rates, such a phenomenon, if it exists, appears to be overwhelmed by the apparent substitutability of home time in the utility and q -functions over the life cycle. The results therefore appear to be more consistent with the notion of a permanent lifetime market participation rate (Mincer 1962), in which current participation is adapted to transitory events such as a “timing” failure (twins) and transitory labor force withdrawal or entry has little impact on the permanent level.

We have so far interpreted the participation results based on the pooled sample in a life-cycle context, even though we have not accounted for the possibility of a vintage-based explanation. As was stressed, the data are not well suited to such an endeavor, though we now report on two further analyses, each based upon the separate surveys to provide limited evidence on the validity of the life-cycle interpretation. First, we disaggregated by survey year and performed the same logit analyses for the life-cycle stages of the two actual cohorts that can be identified from the pooled data, as (1) the 1965, 15–24 cohort corresponds roughly to the 1973, 25–34, age at first birth under 25 group; and (2) the 1965, 25–34 cohort corresponds to

the 1973, 35–44 age group with age at first birth under 35. The results, presented in table 4, confirm the existence of a declining twins-first effect with age for the same cohort—indicated by arrows—although statistical significance at conventional levels is impossible to achieve, given the very few twins-first occurrences.

The second piece of confirmatory evidence, obtained from information contained only in the 1973 survey, is based upon the fraction of potential years since the birth of the first child spent by the mother in market work. The same twins-first regression scheme indicated that women aged 35–44 who experienced a twins-first birth actually worked a slightly greater fraction of their potential work experience, spending about .75 of a year more in the labor force (out of a potential experience of over 19 years and an average actual work experience of over 6 years). Again, the initial increase in home time due to a twins birth that is indicated in the 15–24 age group appears to have had no dominant permanent influence on future home time.

c) *Exogenous and Endogenous Fertility Effects on Participation:
Estimates of the Bias*

The twins-first results in general confirm the existence of an important causal connection between fertility (the price of children) and labor supply that has ramifications throughout the life cycle. As noted, however, most of the female labor supply literature has assumed actual fertility to be “exogenous.” We now use the twins-first results to test whether the conditions hold under which the *ceteris paribus* relationship between *observed* fertility and labor supply provides unbiased estimates of price-effect information. This is done by estimating the difference between the truly exogenous effects on labor supply provided by *TF* and those effects implied by the relation-

TABLE 4
IMPACT OF TWINS-FIRST BIRTH ON LIFE-CYCLE LABOR FORCE
PARTICIPATION: “TRUE” COHORTS FROM SEPARATE SAMPLES

Sample	15–24	25–34	35–44	Estimation Technique
1965	–.224 (.132)	–.229 (.168)	...	OLS*
1973 (<i>AFB</i> < 25)	...	–.028 (.117)	...	ML LOGIT
1973 (<i>AFB</i> < 35)	...	...	.043 (.100)	ML LOGIT

NOTE.—From left to right there are 7, 7, 18, and 25 twins-first observations, respectively.

*These are OLS regressions, since the *TF* = 1, *NOW* = 1 cell was empty. For comparison, the corresponding OLS estimates for table 3 were –.226 (.114) and –.087 (.088).

ships among *CEB*, *CHILD*, and participation, that is, the bias in the fertility-participation relationship caused by heterogeneity in tastes.

It is a relatively simple matter to address the bias question once a labor supply equation is fully specified, since the extent of the bias will, in general, depend upon what is relegated to the error term. We therefore experimented with several versions of the participation function and found the estimated conventional fertility measure (*CEB* and *CHILD*) relationships with participation to be robust to the inclusion of a variety of variables. For this reason, we report on the simplest version only, namely, logits which include solely *CEB* and *CHILD*. The method for calculating the bias is straightforward: estimate the following participation equation,

$$NOW = \gamma_1 CEB + \gamma_2 CHILD + \epsilon_1,$$

and compute

$$\frac{dNOW}{dTF} = \hat{\gamma}_1 \frac{\partial CEB}{\partial TF} + \hat{\gamma}_2 \frac{\partial CHILD}{\partial TF},$$

where $\partial CEB/\partial TF$ and $\partial CHILD/\partial TF$ are obtained from table 2, and $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are transformed logit coefficients. This biased estimate of the exogenous impact of twins first, denoted by $(dNOW/dTF)_B$, can then be compared with the unbiased estimate contained in table 3, $(dNOW/dTF)_U$. These figures are summarized in table 5.

It is evident that the fertility effect, that is, the combined *CEB* and *CHILD* effect, is substantially dampened at all ages under the (erroneous) assumption that actual fertility does not reflect differences in preference orderings. While the cause of this bias in the *TF* impact cannot be assigned to *CEB* or *CHILD* individually—one cannot tell whether the effect of younger children or that of older children is biased, nor in which direction—their combined effect is grossly underestimated in absolute value. The covariation between actual fertility

TABLE 5
COMPARISONS OF EXOGENOUS AND ACTUAL FERTILITY
"EFFECTS" ON LIFE-CYCLE PARTICIPATION,
POOLED SAMPLE

	AGE GROUP		
	15-24	25-34 (<i>AFB</i> < 25)	35-44 (<i>AFB</i> < 25)
$\left(\frac{dNOW}{dTF} \right)_B$	-.103	.024	.041
$\left(\frac{dNOW}{dTF} \right)_U$	-.371	-.102	+.142

and participation estimated from a cross section of women therefore understates significantly the extent to which female labor supply responds to changes in the price of fertility control (p_n) relative to the responsiveness of fertility to alterations in contraceptive costs. Moreover, these results imply that estimates of conditional *own*-price (wage) and income effects in labor supply equations which include a measure of actual fertility among the regressors will be inconsistent. While it is possible that these inconsistent estimates of “short-run” price and income effects conditioned on the quantity of children will be fortuitously close to consistent estimates of such effects forthcoming from the usual demand functions (which provide the information appropriate to the Slutsky-type of relations implied by theory), such fertility-conditioned estimates may stray far from those required to either test theory or draw policy conclusions.

V. Conclusions

In the first section of this paper we explored a two-period labor supply model which incorporated production functions for child quality and human capital and in which individuals differed in their preference orderings for leisure and/or children. The conditions under which, given lack of information on the price of children (e.g., the price of contraception) and preference orderings, labor supply equations that include children as exogenous regressors will not confound tastes and price heterogeneity were derived. It was demonstrated that if the price of children were known, the use of price as a regressor dominates the use of quantity, as simultaneous-equations estimates were shown to provide conditional demand estimates which must rely on the existence of price information to exogenously alter quantities independently of tastes. Empirical models of the simultaneous determination of fertility and labor supply are thus no more informative than the usual set of consumer demand equation estimates (to which they must correspond) in terms of verifying theory.

In Section II, we examined the informational content of fertility effects. Within a rationing (of children) framework, it was shown that the sign of the compensated effect of a change in the price of children on home time provided by the impact of additional (exogenous) children on labor supply could provide information on the existence of the unobserved child quality production function and human capital depreciation only if restrictions on utility-function parameters are imposed. Given strong separability in utility, however, only a weak test for the existence of a child quality dimension was derived based on a decomposition of cross-price effects.

To simulate exogenous price variation, we developed and im-

plemented a methodology based on the natural event of a twins birth. In particular, we traced the impact of a twins occurrence in the first pregnancy on subsequent fertility and labor supply behavior as well as on contemporaneous female participation. These results were explicitly related to the life-cycle model discussed in previous sections. Two important conclusions emerged, qualified importantly by our small sample of twins-first births. First, the use of actual fertility in participation equations greatly understates the impact of exogenous changes in fertility on female work status. Second, total fertility and measures of lifetime labor supply appear to be unaffected by the transitory increase in children experienced by women with twins in the first birth. Initial withdrawal from market work, which was shown to be supportive of the time-intensive nature of child quality investment, is compensated almost exactly by the earlier reentrance of women later in the life cycle. The twins-first labor supply effects obtained, however, suggest that reductions in contraceptive costs, which induce a decline in total fertility, play a causative role in increasing female participation.

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